ENTERPRISE DATA GOVERNANCE FRAMEWORK DESIGN & IMPLEMENTATION FOR A GLOBAL MANUFACTURING COMPANY

DATA GOVERNANCE TARGET OPERATING MODEL (TOM)

1. EXECUTIVE SUMMARY

The Data Governance Target Operating Model (TOM) defines the structure, roles, processes, and technologies required to manage data as a strategic asset across the enterprise. It ensures alignment between business objectives, regulatory requirements, and data management practices.

2. OBJECTIVE

Establish an enterprise-wide Target Operating Model (TOM) that defines governance structure, organizational roles, operating processes, technology enablement, and performance management to ensure trusted, standardized, and high-quality data across all manufacturing operations, supply chain functions, and corporate business units.

3. OPERATING MODEL OVERVIEW

3.1. Governance Model Type

✓ **Federated Governance Model**

- Central governance defines policies & standards
- Domain teams execute governance across business domains and manufacturing sites

The Target Operating Model is designed based on a Federated Data Governance Model, where governance policies, standards, and oversight are centralized, while execution and accountability are distributed across business domains and manufacturing sites.

✓ **Recommended for a Global Manufacturing Company because:**

A Global Manufacturing Company typically consists of:

- Multiple manufacturing plants
- Multiple countries or regions
- Shared corporate functions
- Multiple ERP instances
- Diverse business processes
- Central corporate governance with local operational execution

A federated model provides:

- Centralized policy and standards
- Decentralized execution
- Local business accountability
- Global consistency

3.2. Design Principles

- Enterprise-wide governance with local accountability
- Business-owned data with IT-enabled platforms
- Standardized policies and processes
- Single Source of Truth (SSOT) for master data
- Data treated as a strategic enterprise asset
- Scalable governance across global operations
- Risk-based and compliance-driven governance
- Data quality embedded into operational processes

4. GOVERNANCE STRUCTURE

4.1. Organizational Layers

1. **Level 1: Board of Directors**

   Responsibilities:

   - Approve enterprise governance strategy
   - Monitor strategic risks
   - Review governance performance

2. **Level 2: Executive Steering Committee**

   Members:

   - CEO
   - COO
   - CFO
   - CIO
   - CDO
   - Chief Supply Chain Officer
   - Manufacturing Director

Responsibilities:

- Sponsor the transformation program
- Resolve cross-functional issues
- Approve investments
- Review KPI performance

3. **Level 3: Data Governance Council**

   Typical Members:

   - Chief Data Officer
   - CIO
   - Manufacturing Director
   - Supply Chain Director
   - Finance Director
   - HR Director
   - Quality Director
   - Compliance Lead
   - Information Security Lead

   Responsibilities:

   - Approve governance policies
   - Approve enterprise standards
   - Resolve cross-domain issues
   - Prioritize governance initiatives
   - Monitor enterprise KPIs

4. **Level 4: Chief Data Officer (CDO)**

   Responsibilities:

   - Own the Data Governance Strategy
   - Lead enterprise implementation
   - Report governance maturity
   - Chair governance initiatives
   - Ensure cross-functional alignment

5. **Level 5: Data Governance Office (DGO)**

   Responsibilities:

   - Develop governance policies
   - Facilitate governance forums
   - Maintain business glossary
   - Manage metadata repository
   - Coordinate stewardship activities
   - Monitor governance KPIs
   - Support audits and compliance

6. **Level 6: Data Owner**

   Accountable for:

   - Data quality
   - Business definitions
   - Access approval
   - Policy compliance
   - Data lifecycle decisions

7. **Level 7: Data Steward**

   Responsibilities:

   - Daily monitoring
   - Data quality validation
   - Metadata maintenance
   - Issue investigation
   - Business glossary updates
   - Rule implementation

8. **Level 8: IT Data Custodian**

   Responsibilities:

   - Security controls
   - Infrastructure
   - Backup & Recovery
   - Access management

9. **Level 9: Data Consumers**

   Examples:

   - Plant Managers
   - Production Supervisors
   - Procurement Analysts
   - Supply Chain Analysts
   - Finance Analysts
   - BI Developers
   - Executives

5. BUSINESS DATA DOMAINS

Data Domain	**Example Data Assets**	**Business Owner**
Manufacturing	Production Orders, BOM, Routing, Work Centers	Manufacturing Director
Supply Chain	Suppliers, Purchase Orders, Inventory	Supply Chain Director
Product	Material Master, Product Specifications	Product Management
Customer	Customer Master, Contracts	Sales Director
Finance	GL Accounts, Cost Centers, Fixed Assets	Finance Director
HR	Employees, Organizational Structure	HR Director
Quality	Inspection Results, Non-Conformance Records	Quality Director
Maintenance	Equipment, Assets, Preventive Maintenance	Maintenance Manager

6. RACI FRAMEWORK

Governance Activity	DGC	CDO	DGO	DO	DS	IT	EA	SEC
Approve Data Governance Policy	A	R	C	I	I	I	I	C
Approve Enterprise Standards	A	R	C	C	I	I	C	C
Define Business Glossary	I	C	R	A	R	I	C	I
Define Critical Data Elements	I	C	R	A	R	I	I	I
Data Quality Rules	I	C	C	A	R	C	I	I
Master Data Governance	I	C	R	A	R	C	C	I
Metadata Management	I	C	A	R	R	C	C	I
Data Quality Monitoring	I	C	C	A	R	C	I	I
Issue Resolution	I	C	C	A	R	C	I	I
KPI Reporting	C	A	R	C	C	I	I	I
Access Approval	I	C	I	A	C	R	I	C
Security Controls	I	I	I	C	I	R	C	A

Legend: R = Responsible, A = Accountable, C = Consulted, I = Informed

7. GOVERNANCE CAPABILITIES

Capability	Description
Data Governance	Policies, standards, decision rights
Data Quality Management	Monitoring and improvement of data quality
Master Data Management	Golden record and master data governance
Metadata Management	Business glossary, catalog, lineage
Reference Data Management	Standardized reference values
Data Security	Classification, access control, encryption
Data Privacy	Protection of personal and sensitive data
Data Lifecycle Management	Creation, retention, archival, deletion
Data Architecture	Enterprise data models and integration
Data Issue Management	Incident handling and remediation

8. CORE GOVERNANCE PROCESSES

8.1. Data Quality Management

- **Objective:**

  Ensure enterprise data is accurate, complete, consistent, timely, and fit for business operations, regulatory compliance, and strategic decision-making.

- **Define Critical Data Elements (CDE)**

  Determination Criteria

  Critical Data Elements (CDEs) are identified based on one or more of the following criteria:

  - Regulatory, financial, or audit reporting impact
  - Manufacturing and production impact
  - Supply chain and procurement impact
  - Customer fulfillment and service impact
  - Product quality and compliance impact
  - Operational dependency across multiple business functions
  - Financial and inventory valuation impact
  - Executive reporting and business intelligence requirements

- **Example Critical Data Elements (CDE)**

Business Domain	Critical Data Element
Material	Material ID
Product	Product Code / SKU
Manufacturing	Production Order Number
Manufacturing	Bill of Materials (BOM)
Manufacturing	Routing
Inventory	Inventory Quantity
Supplier	Supplier ID
Customer	Customer ID
Logistics	Warehouse Location
Quality	Quality Inspection Result

Business Domain	Critical Data Element
Equipment	Asset / Equipment ID
Finance	Cost Center
Plant	Plant Code

- **Data Quality Dimensions**
  - Accuracy
  - Completeness
  - Consistency
  - Validity
  - Timeliness
  - Uniqueness
  - Integrity
- **Data Quality Monitoring**

  Activities:
  - Define enterprise Data Quality Rules
  - Establish Data Quality KPIs and thresholds
  - Perform automated data profiling
  - Monitor data quality dashboards
  - Review quality trends
  - Escalate critical issues

Enterprise Data Quality KPIs:

KPI	Definition	Target	Reporting Frequency
Overall Data Quality Score	Composite score across all quality dimensions	**≥95%**	Monthly
Accuracy	Records accurately reflect real-world business data	**≥98%**	Monthly
Completeness	Mandatory attributes are populated	**≥98%**	Weekly

KPI	Definition	Target	Reporting Frequency
Consistency	Data values are consistent across integrated systems	**≥98%**	Weekly
Validity	Data complies with business rules and reference values	**≥99%**	Weekly
Uniqueness	No duplicate master data records	**≥99.5%**	Monthly
Timeliness	Data is available within the defined SLA	**≥95%**	Daily
Integrity	Relationships between related datasets remain valid	**≥99%**	Weekly

- Data Issue Management Workflow:
  - Detect → Log → Assign → Investigate → Root Cause Analysis → Resolve → Validate → Close → Continuous Improvement

8.2. Metadata Management

- **Objective:**

  Provide a common understanding of enterprise data through standardized metadata and business definitions.

- **Core Activities**
  - Maintain Enterprise Data Catalog
  - Maintain Business Glossary
  - Maintain Data Dictionary
  - Register Technical Metadata
  - Document Data Owners
  - Maintain Data Lineage
  - Track Metadata Changes
  - Periodically review metadata completeness

- **Metadata Components**

Component	**Description**
Business Glossary	Standard business definitions
Data Dictionary	Technical data definitions
Metadata Repository	Enterprise metadata inventory
Data Catalog	Searchable enterprise data assets
Data Lineage	Source → Transformation → Consumption
Data Ownership	Data Owner and Steward assignments

- **Data Lineage**

Supplier Portal

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ERP

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MES

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Data Integration

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Enterprise Data Warehouse

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Power BI

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Executive Dashboard

8.3. Data Access & Security

- **Objective:**

  Protect enterprise information assets while enabling secure and authorized access.

- **Access Management**
  - Mandatory Role-Based Access Control (RBAC)
  - Attribute-Based Access Control (ABAC) for advanced scenarios
  - Least Privilege Principle
  - Segregation of Duties (SoD)
  - Need-to-Know Principle

- **Access Workflow**

  User Request

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  Manager Approval

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  Data Owner Approval

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  IT Security Validation

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  Access Provisioning

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  Audit Logging

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  Quarterly Access Review

- **Periodic Review**
  - Quarterly user access review
  - Annual privileged account review
  - Immediate revocation for terminated employees
  - Review after organizational changes

8.4. Data Lifecycle Management

- **Define lifecycle stages**:
  - Create → Store → Use → Share → Archive → Retain → Secure Delete
- **Lifecycle Activities**

Stage	Activities
Create	Data creation and validation
Store	Secure storage
Use	Business operations
Share	Controlled data sharing
Archive	Long-term retention
Retain	Regulatory retention
Delete	Secure disposal

- **Retention policies based on regulatory requirements**

  Retention Policy (Illustrative)

Data Type	Retention Period*
Production Records	7 years
Quality Inspection Records	10 years
Material Master	Active + 5 years
Supplier Master	Active + 7 years
Customer Master	Active + 5 years
Financial Records	7 years
Maintenance Records	10 years
System Logs	1–2 years

**Illustrative defaults. Actual retention periods should be validated with Legal, Compliance, Finance, and applicable regulations in each operating jurisdiction.*

8.5. Data Classification

- Define classification levels:
    - Public
    - Internal
    - Confidential
    - Restricted (PII / highly sensitive)

9. TECHNOLOGY ENABLEMENT

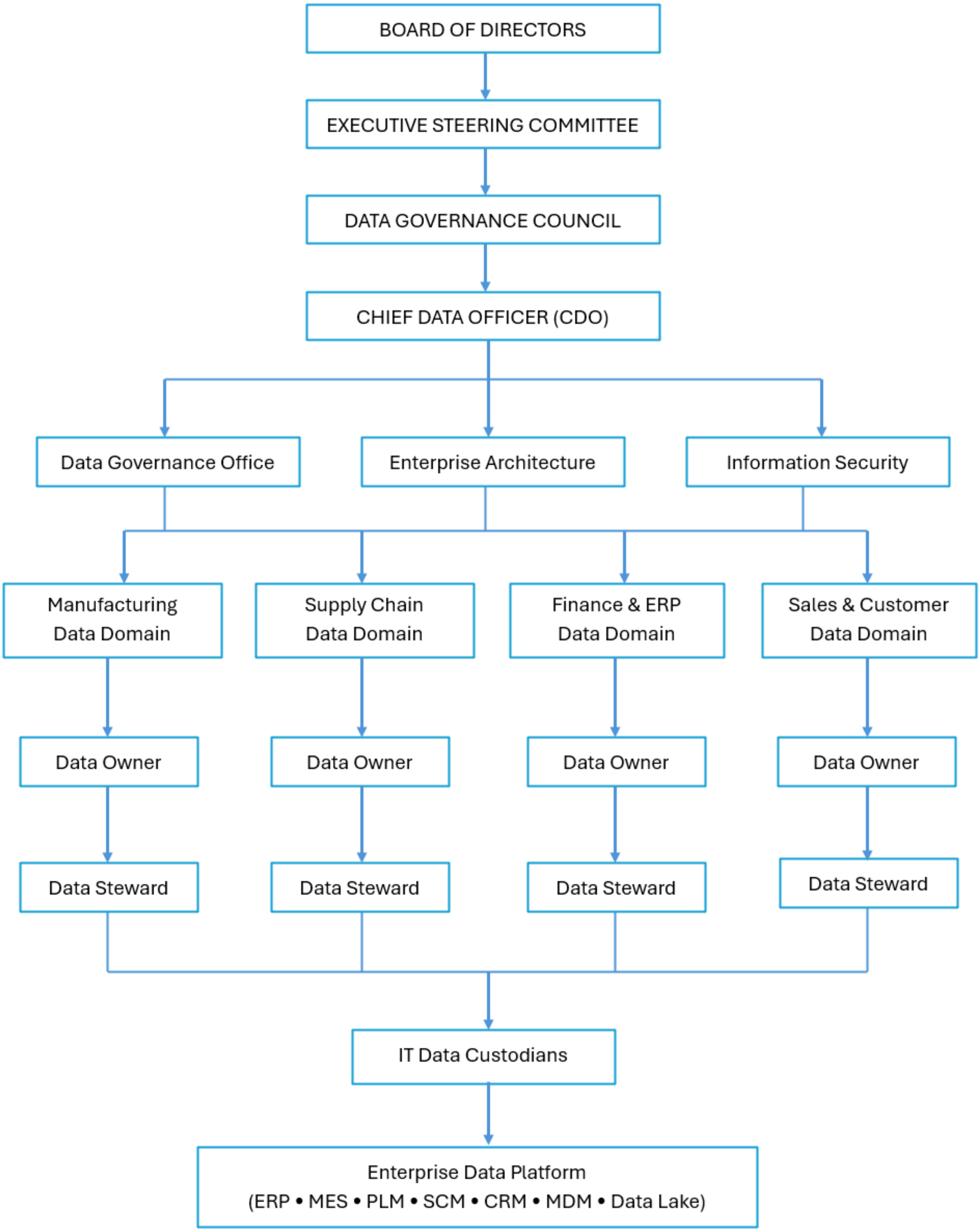
Source Systems:

- ERP (SAP S/4HANA / Oracle ERP Cloud / Microsoft Dynamics 365)
- MES (Manufacturing Execution System)
- PLM (Product Lifecycle Management)
- SCM (Supply Chain Management)
- CRM
- Warehouse Management System (WMS)
- IoT Sensors
- Quality Management System (QMS)

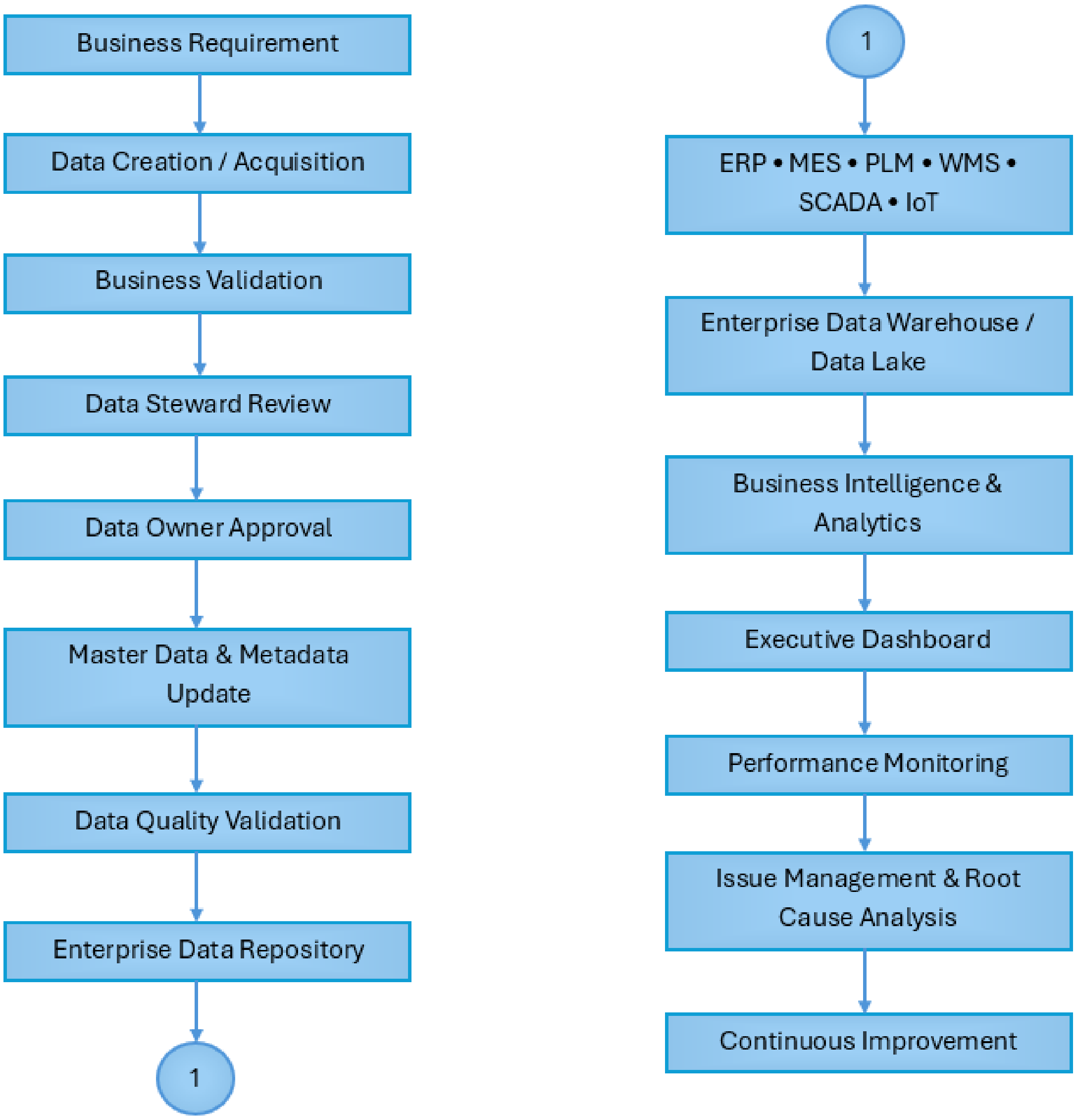
Enterprise Data Platform:

- Enterprise Data Warehouse
- Data Lake
- Data Catalog
- Metadata Repository
- MDM Platform
- Data Quality Engine
- Data Integration Platform
- Business Intelligence Platform

10. DATA GOVERNANCE ARCHITECTURE



11. GOVERNANCE FLOW DIAGRAM



12. DATA GOVERNANCE FRAMEWORK COMPONENTS

12.1. Policy Layer

- **Objective**

  Define enterprise-wide governance policies, standards, and controls that ensure data is managed consistently throughout its lifecycle.

- **Core Policies**

Policy	Purpose
Enterprise Data Governance Policy	Defines governance principles, operating model, roles, responsibilities, and decision rights.
Data Quality Policy	Establishes enterprise data quality dimensions, rules, monitoring, remediation, and KPIs.
Metadata Management Policy	Defines standards for business glossary, data catalog, metadata repository, and data lineage.
Master Data Management (MDM) Policy	Governs the creation, maintenance, synchronization, and governance of master data across enterprise systems.
Data Classification Policy	Defines data classification levels and handling requirements based on business sensitivity.
Data Access & Security Policy	Establishes access control, authentication, authorization, encryption, and protection of enterprise information assets.
Data Lifecycle Management Policy	Defines data creation, storage, usage, archival, retention, and secure disposal requirements.
Data Issue & Incident Management Policy	Establishes processes for identifying, resolving, escalating, and preventing data quality and governance issues.

- **Key Deliverables**
  - Enterprise Governance Standards
  - Governance Procedures
  - Data Standards
  - Standard Operating Procedures (SOPs)
  - Compliance Guidelines

12.2. Process Layer

- **Objective**

  Standardize governance processes that ensure enterprise data remains accurate, consistent, secure, and fit for business use.

- **Core Governance Processes**

Process	Description
Data Ownership	Assign accountability for Critical Data Elements (CDEs), business rules, approvals, and governance decisions.
Data Stewardship	Manage day-to-day data quality, metadata, issue resolution, and business rule compliance.
Master Data Management (MDM)	Maintain Golden Records, eliminate duplicates, and synchronize master data across ERP, MES, PLM, WMS, and other enterprise systems.
Data Quality Management	Define data quality rules, perform profiling, monitor KPIs, and execute corrective actions.
Metadata Management	Maintain the business glossary, data catalog, metadata repository, and end-to-end data lineage.
Data Issue Management	Detect, log, prioritize, investigate, resolve, validate, and close data-related issues within defined SLAs.
Data Security & Privacy	Apply access controls, data classification, encryption, monitoring, and privacy controls.
Data Lifecycle Management	Govern data from creation through archival and secure deletion in accordance with business and regulatory requirements.

12.3. Technology Layer

- **Objective**

  Provide enterprise technology capabilities that automate governance, improve data quality, and support trusted analytics.

- **Core Technologies**

Technology	Primary Purpose
Enterprise Data Catalog	Enables users to discover, understand, and access governed enterprise data assets.
Metadata Repository	Stores business, technical, and operational metadata in a centralized repository.
Data Quality (DQ) Platform	Profiles data, executes validation rules, monitors quality metrics, and generates alerts.
Master Data Management (MDM) Platform	Creates and maintains Golden Records while synchronizing master data across enterprise applications.
Data Lineage Tool	Documents end-to-end data flow from source systems through transformations to reporting and analytics.
Enterprise Data Warehouse (EDW)	Consolidates governed enterprise data for standardized reporting and analytics.
Data Lake Platform	Stores structured and unstructured manufacturing and operational data for advanced analytics.
Business Intelligence Platform	Provides dashboards, KPI monitoring, executive reporting, and self-service analytics.

- **Integrated Enterprise Systems**
  - ERP
  - MES
  - SCADA
  - Industrial IoT
  - PLM
  - SCM
  - WMS
  - CRM
  - Quality Management System (QMS)
  - Enterprise Asset Management (EAM)
  - Data Warehouse
  - Data Lake

12.4. People Layer

- **Objective**

  Establish clear governance roles and accountability to ensure effective enterprise data management across business and IT.

Role	**Primary Responsibilities**
Chief Data Officer (CDO)	Defines enterprise data strategy, sponsors governance initiatives, and reports governance performance to executive leadership.
Data Governance Office (DGO)	Coordinates governance activities, maintains policies and standards, facilitates governance forums, monitors KPIs, and drives continuous improvement.
Data Owners	Own critical business data, approve standards, define business rules, authorize access, and are accountable for data quality within their domains.
Data Stewards	Execute day-to-day governance activities, maintain metadata, monitor data quality, resolve issues, and support business users.
IT Data Custodians	Implement technical controls, manage enterprise platforms, ensure system availability, maintain integrations, backups, security, and infrastructure supporting governed data.

12.5. Expected Outcomes

Framework Component	**Business Outcome**
Policy Layer	Consistent governance standards, regulatory compliance, and standardized data management practices.
Process Layer	Clear accountability, standardized workflows, improved data quality, and efficient issue resolution.
Technology Layer	Automated governance capabilities, trusted metadata, master data synchronization, and reliable analytics.
People Layer	Strong governance leadership, defined ownership, cross-functional collaboration, and sustainable governance adoption.

13. COMPLIANCE & RISK MANAGEMENT

13.1. Compliance Management

Key Activities

- Align Data Governance policies with applicable **data protection, cybersecurity, financial reporting, industry, and regional regulatory requirements** (e.g., GDPR principles where applicable).
- Establish and maintain enterprise-wide **Data Governance Policies, Standards, and Standard Operating Procedures (SOPs)**.
- Implement **end-to-end audit trails** for critical data creation, modification, approval, and deletion activities.
- Perform **periodic compliance assessments** to evaluate adherence to governance policies and regulatory obligations.
- Conduct **internal governance audits** and support external regulatory or customer audits.
- Monitor compliance with **Data Retention, Data Classification, Access Control, and Data Quality** policies.
- Maintain evidence and documentation to demonstrate governance compliance during audits.
- Track compliance findings and implement corrective actions through a structured remediation process.

13.2. Risk Management

Key Risk Management Activities

- Identify enterprise data risks across business processes, systems, and third-party integrations.
- Assess risks based on **business impact** and **likelihood**.
- Define preventive, detective, and corrective controls.
- Maintain an enterprise **Data Risk Register**.
- Monitor key risk indicators (KRIs) and governance KPIs.
- Escalate high-risk issues to the **Data Governance Council** and Executive Steering Committee.
- Review risks periodically and after significant business or technology changes.

13.3. Key Compliance Controls

- Enterprise Data Governance Policies and Standards
- Data Classification and Handling Controls
- Role-Based Access Control (RBAC)
- Multi-Factor Authentication (MFA)
- Data Encryption (at rest and in transit)
- Data Retention and Secure Disposal
- Audit Logging and Monitoring
- Periodic Access Reviews
- Data Quality Monitoring
- Master Data Governance Controls
- Change Management and Approval Workflow
- Internal Audit and Compliance Reviews

13.4. Governance Compliance Monitoring

Activity	Frequency	Owner
Data Quality KPI Review	Monthly	Data Governance Office (DGO)
Data Owner Certification	Quarterly	Data Owners
User Access Review	Quarterly	IT Security & Data Owners
Policy Compliance Assessment	Semi-Annually	Data Governance Office
Internal Data Governance Audit	Annually	Internal Audit
Governance Maturity Assessment	Annually	Data Governance Council
Enterprise Risk Review	Quarterly	Enterprise Risk Management & DGO

14. KPIs & SUCCESS METRICS

KPI	Before (Current State)	Target (Future State)	Success Criteria
Enterprise Data Quality Score	~78%	**≥95%**	Enterprise data consistently meets quality thresholds across all critical data domains.
Critical Data Elements (CDEs) with Assigned Data Owner	25%	**100%**	Every Critical Data Element has a formally assigned Data Owner accountable for governance.
Business Domains with Assigned Data Steward	30%	**100%**	All business domains have designated Data Stewards actively performing governance activities.
Metadata Coverage	20%	**≥95%**	Critical enterprise data assets are documented in the enterprise metadata repository and data catalog.
Business Glossary Completion	15%	**100%**	Enterprise business terms are standardized, approved, and published across all business functions.
Master Data Golden Record Coverage	45%	**≥98%**	Master data entities (Material, Supplier, Customer, Product, Plant) are managed as Golden Records.
Duplicate Material Records	8%	**<0.5%**	Duplicate material records are minimized through MDM governance and automated matching.
Duplicate Supplier Records	6%	**<0.5%**	Supplier master data duplication is effectively eliminated.
Duplicate Customer Records	7%	**<1%**	Customer master data maintains high uniqueness and consistency.
Data Quality Rule Coverage	35%	**100%**	All Critical Data Elements have documented and implemented Data Quality Rules.
Automated Data Quality Monitoring	20%	**100%**	Enterprise Data Quality monitoring is automated for all critical data domains.
Data Issue SLA Compliance	65%	**≥95%**	Data quality issues are resolved within agreed service-level targets.
Critical Data Issues Open >30 Days	18	**0**	No unresolved critical data issues exceed 30 calendar days.
Policy Compliance Rate	60%	**≥98%**	Business units consistently comply with approved Data Governance Policies.
Quarterly Data Steward Review Completion	40%	**100%**	All scheduled stewardship reviews are completed on time.
Quarterly Data Owner Certification	Not Performed	**100%**	Data Owners formally certify data ownership, quality, and compliance each quarter.
Enterprise Data Catalog Adoption	10%	**≥90%**	Business users actively use the enterprise data catalog to discover and understand data assets.
Data Lineage Coverage	5%	**≥90%**	End-to-end lineage is documented for critical reports and data assets.
Sensitive Data Classification Coverage	40%	**100%**	All sensitive and confidential data assets are classified according to enterprise policy.
Access Review Completion	55%	**100%**	Quarterly access reviews are completed for all governed systems and sensitive data.
High Severity Data Security Incidents	5/year	**0**	No critical data security or privacy incidents attributable to governance failures.
Audit Findings Related to Data Governance	12 Major Findings	**0 Major Findings**	Internal and external audits report no major governance deficiencies.
ERP–MES Master Data Synchronization Success Rate	82%	**≥99%**	Master data synchronization between enterprise systems is reliable and consistent.
Manufacturing Master Data Accuracy	88%	**≥99%**	Material, BOM, Routing, and Plant master data accurately reflect approved business records.
Executive Report Consistency	70%	**100%**	Executive reports are generated from a single governed source with consistent definitions.
Time to Identify Data Issues	5 days	**<4 hours**	Critical data issues are detected rapidly through automated monitoring.
Time to Resolve Critical Data Issues	15 days	**<24 hours**	Critical data issues are resolved within established governance SLAs.
Self-Service Analytics Adoption	20%	**≥80%**	Business users confidently access governed data for analytics without IT intervention.
Business User Trust in Enterprise Data	60%	**≥95%**	Survey results indicate high confidence in enterprise data quality and reliability.
Enterprise Governance Maturity (DAMA/DCAM-aligned)	Level 2 (Developing)	**Level 4 (Managed)**	Governance capabilities are standardized, measured, and continuously improved across the enterprise.

15. IMPLEMENTATION ROADMAP

15.1. Executive Roadmap Overview

Phase	Timeline	Primary Objective
Phase 1	Months 1–3	Governance Foundation
Phase 2	Months 4–6	Governance Standardization
Phase 3	Months 7–12	Technology Enablement & Enterprise Rollout
Phase 4	Months 13–18	Optimization & Continuous Improvement

15.2. Implementation Roadmap

1. Phase 1 – Governance Foundation (Months 1–3)

Objective:

Establish the organizational, policy, and governance foundations required to manage enterprise data consistently across all manufacturing business domains.

Deliverables:

- Governance Charter
- Governance Organization
- Operating Model
- Policy Framework
- Governance Roles
- Current State Report
- Gap Assessment
- Data Domain Catalogue

2. Phase 2 – Governance Standardization (Months 4–6)

Objective:

Standardize enterprise data management processes and governance capabilities.

Deliverables:

- Enterprise Business Glossary
- Metadata Repository
- Data Catalog
- Data Quality Rules

- MDM Standards
- Governance Procedures
- Standard Operating Procedures (SOPs)

3. **Phase 3 – Technology Enablement & Enterprise Rollout (Months 7–12)**

Objective:

Deploy governance technologies, integrate enterprise systems, and operationalize governance across the organization.

Deliverables:

- Production-ready Governance Platform
- Data Catalog
- MDM Hub
- Automated Data Quality Monitoring
- KPI Dashboard
- Governance Training Materials
- Change Management Plan

4. **Phase 4 – Optimization & Continuous Improvement (Months 13–18)**

Objective:

Enhance governance maturity through continuous monitoring, optimization, automation, and advanced analytics.

Deliverables

- Governance Performance Dashboard
- Continuous Improvement Register
- Updated Governance Policies
- AI-enabled Data Quality Capabilities
- Governance Maturity Assessment Report
- Executive Performance Report

15.3. Major Milestones

Milestone	Target Completion
Governance Organization Established	Month 1
Data Governance Policies Approved	Month 3
Data Owners & Data Stewards Assigned	Month 3
Business Glossary Published	Month 5
Enterprise Metadata Repository Operational	Month 6
Enterprise Data Catalog Deployed	Month 8
MDM Platform Operational	Month 9
Data Quality Monitoring Automated	Month 10
ERP–MES–SCADA Integration Governed	Month 11
Executive KPI Dashboard Live	Month 12
Governance Maturity Reassessment	Month 18

16. CHANGE MANAGEMENT & ADOPTION

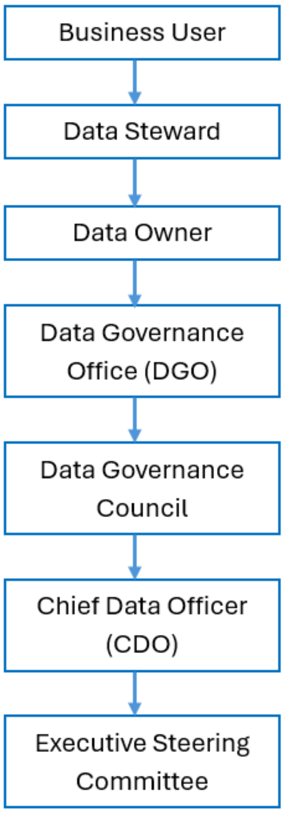
- Stakeholder engagement plan
- Training & awareness programs
- Governance champions per domain
- Communication strategy

17. ESCALATION FRAMEWORK

17.1. Escalation Principles

- The Data Governance Escalation Framework ensures that:
- Data issues are resolved at the lowest appropriate organizational level whenever possible.
- Clearly defined ownership and accountability exist throughout the escalation process.
- High-impact issues affecting multiple business functions are escalated promptly.
- Service Level Agreements (SLAs) are monitored and enforced.
- Root Cause Analysis (RCA) and corrective actions are completed before issue closure.
- Lessons learned are incorporated into continuous improvement initiatives.

17.2. Escalation Hierarchy



17.3. Severity Classification

Severity	Business Impact	Escalation Level
Critical	Production disruption, regulatory breach, financial reporting error, cybersecurity incident, or enterprise-wide data failure	CDO → Executive Steering Committee
High	Multiple business functions or manufacturing plants affected	Data Governance Council
Medium	Single business function affected with moderate operational impact	Data Governance Office
Low	Localized issue with minimal business impact	Data Owner / Data Steward

17.4. Service Level Agreement (SLA)

Service Level Agreement (SLA) (Example):

Severity	Initial Response	Target Resolution	Escalation Level
Critical	< 4 hours	< 24 hours	CDO / Executive Steering Committee
High	< 1 business day	< 5 business days	Data Governance Council
Medium	< 2 business days	< 10 business days	Data Governance Office
Low	< 5 business days	< 30 business days	Data Owner / Data Steward

18. REVIEW & CONTINUOUS IMPROVEMENT

- Quarterly governance review
- Annual TOM update
- Continuous KPI monitoring

19. APPENDICES

Appendix A: Data Quality Dimensions

- Accuracy
- Completeness
- Consistency
- Validity
- Timeliness
- Uniqueness
- Integrity

Appendix B: Sample Workflow

- Data Issue Management Workflow Diagram